

A growing epithelial spheroid in an extracellular matrix: the Plexus2 implementation

prototype/ecm · runs in log/okuda_ECM · 6 August 2026

1. Sets, operators, schedule

A Plexus model is sets $\times$ fields $\times$ operators $\times$ schedule. The biological hierarchy here is four entities and one field; the basement membrane is a *specialised* ECM, not the ECM:

entity / set	representation	principal operators
epithelium vertex, cell	3D active vertex model	seed_mesh_3d build the shell cell_geometry_3d area, perimeter, volume per cell shape_energy_3d the mechanics morphogen_growth_3d grow each cell's target divide_3d divide on volume doubling ecm_growth_gate_3d matrix stress slows the cycle
junction a relation, not a set	half-edges of the mesh, state keyed by vertex pair	junction_myosin per-junction myosin reconnect_t1_3d neighbour exchange (T1)
basement membrane basement_membrane_particle	nodes + explicit crosslinks	seed_basement_membrane lay the sheet down basement_membrane_bond the crosslink force basement_membrane_remodel rest lengths creep basement_membrane_bond_break fragmentation basement_membrane_secrete add material as it grows integrin_adhesion tether to the surface
surface surface	one node per membrane node: direction, radius	surface_track write the recorded boundary
interstitial ECM mpm_particle	MPM fibres	seed_ecm fibres, alignment, density cell_to_ecm the epithelium's push cell_exclude_3d non-penetration backstop ecm_stress colour by von Mises stress
(field) mpm_grid	MLS-MPM background grid	mpm_scatter / mpm_grid_update / mpm_gather the solve

2. The spheroid: 3D active vertex model

Cells are faces of a closed triangulated shell; cell f 's volume is the wedge from the tissue centre, $v_f = \frac{1}{3} \mathbf{c}_f \cdot \mathbf{N}_f$. The energy is

$$E = \sum_f \left[K_A (A_f - A_f^0)^2 + K_P (P_f - P_f^0)^2 + \frac{1}{2} \Gamma P_f^2 + K_V (v_f - v_f^0)^2 \right] + \Lambda \sum_e m_e \ell_e + K_R \sum_i (|\mathbf{x}_i| - R_0)^2, \quad (1)$$

relaxed overdamped each frame, $\mathbf{x} \leftarrow \mathbf{x} - \eta \mu \nabla E$ with a per-step cap. The factor m_e is new (§5); with $m_e \equiv 1$ Eq. (1) is the stock Okuda energy and every prior run is bit-identical. Growth scales each cell's targets by a cumulative s_f ,

$$s_f \leftarrow s_f (1 + \lambda(\rho + \text{Hill}(a_f))), \quad A_f^0 = A_f^{0,\text{init}} s_f^2, \quad P_f^0 = P_f^{0,\text{init}} s_f, \quad v_f^0 = v_f^{0,\text{init}} s_f^3, \quad (2)$$

and **divide_3d** splits a cell once v_f passes twice its reference. From 200 cells the tissue reaches ~ 6000 and its apical radius $4.66 \rightarrow 16.6$ (arbitrary units) in 402 frames.

3. The ECM: MLS-MPM

The matrix is a material, not a set of springs, so it is solved the way materials are: as a cloud of points that carry the material and a background grid that does the physics. Each frame the points hand their mass and momentum to the grid nodes nearest them, the grid resolves everything arriving at each node into one velocity, and the points read that velocity back and move. Nothing is ever computed between two points directly — which is why bodies that never touch can still push on one another, and why the grid spacing sets the finest thing the matrix can feel.

We seed it as fibres rather than a uniform block, because a fibrous matrix and a homogeneous one respond differently to being pushed, and the interesting behaviour — splaying, dragging, a stress front travelling outward — lives in the fibres.

4. Spheroid $\leftrightarrow$ ECM

The two solvers cannot share a world: the matrix grid is fixed to the unit cube, the epithelium lives in a box fifty times larger, and shrinking the tissue to fit changes which term of its energy dominates and stops it growing. So they are run as two passes joined by one geometric scale, and what crosses between them is a *recorded shape* rather than a live tissue.

That recording needs somewhere to live, and this is what the **surface** Level is. It is a set of nodes, one per membrane node, each holding a fixed direction from the tissue centre and the radius the epithelium reached in that direction at that moment. It is not a mesh and carries no mechanics; it is the epithelium's outer boundary, written down. Before it existed the same information was a lookup table, which meant nothing could act on it and nothing could be attached to it — promoting it to a Level is what lets other things bind to the tissue surface as an object rather than consult it as data.

Everything outside the epithelium then reads that one surface. Matrix fibres that end up inside it are pushed back out, first softly and then, as a backstop, by being placed outside:

$$\mathbf{a}_i = k_c (R(\hat{\mathbf{u}}_i, t) - r_i)_+ \hat{\mathbf{u}}_i, \quad (3)$$

and the force the fibres push back with is collected by direction and divided by the area each direction covers, giving a pressure over the surface. A cell facing a stressed matrix then grows more slowly:

$$g = f + \frac{1-f}{1 + (P/p_{1/2})^n}, \quad s_{\text{now}} \leftarrow s_{\text{prev}} (1 + g(f_{\text{grow}} - 1)). \quad (4)$$

Because it is the growth *rate* that is slowed and not the size directly, a small difference compounds over hundreds of frames — which is how a modest difference in stress between one direction and another can become a difference in shape.

The coupling still runs one way. The surface is a recording, so it drives the matrix and the membrane and neither reaches back, and the second pass cannot change the tissue it is reacting to. Making it mutual means letting an operator write to **surface** instead of only reading it, which is now a change to one Level rather than a change to the architecture.

5. The basement membrane

A basement membrane is a thin sheet of protein an epithelium lays down underneath itself and sits on. It has to do two things at once that pull against each other: hold together as a sheet, and keep covering a surface that grows. We model the load-bearing part of it — the crosslinked collagen IV network — as nodes joined by springs.

Three rules. A crosslink is a spring that slowly *forgets* being stretched and snaps if stretched too far; a node is tied to a fixed *direction* on the surface, which is what makes it adhesion rather than contact, since a node free to slide to the nearest point would never be stretched at all; and everything moves at a speed set by force against drag, with no mass:

$$\gamma \dot{\mathbf{x}}_i = \sum_{j \sim i} \mathbf{f}_{ij} + \mathbf{f}_i^{\text{adh}}, \quad \mathbf{f}_{ij} = k_b (L_{ij} - \ell_{ij}^0) \hat{\mathbf{d}}_{ij}, \quad \dot{\ell}_{ij}^0 = (L_{ij} - \ell_{ij}^0)/\tau_r, \quad (5)$$

$$\mathbf{f}_i^{\text{adh}} = \kappa (\mathbf{a}_i - \mathbf{x}_i), \quad \mathbf{a}_i = \mathbf{c} + \hat{\mathbf{u}}_i (R(\hat{\mathbf{u}}_i, t) + \delta), \quad \hat{\mathbf{u}}_i \text{ frozen at seeding}, \quad (6)$$

with a crosslink dying once $(L - \ell^0)/\ell^0 > \varepsilon_b$.

Forgetting is not optional: a sheet that only stretches cannot enclose a surface that more than triples in size. And because area grows as the square of the radius while new material is added as a fraction of what is already there, the sheet must be *secreted* as the tissue grows or it thins.

Two consequences of having no mass. Only the ratio k_b/γ appears in Eq. (5) — a relaxation *rate* — so a stiffness on its own has no meaning here, and the question worth asking is whether the sheet relaxes faster than the tissue grows. The same ratio bounds what a fixed time step can carry, $\Delta t (z k_b + \kappa)/\gamma < 2$ for z crosslinks meeting at a node, which is about twenty times the bound an inertial sheet would have. Raising γ moves it further, at the price of a sheet that responds more slowly.

Finally, what is called adhesion here is a tether to a surface, not a receptor: the dominant anchor in a real epithelium is to laminin, which is not modelled, and there is no attachment to the surrounding stroma either, so this tether is the only thing holding the sheet.

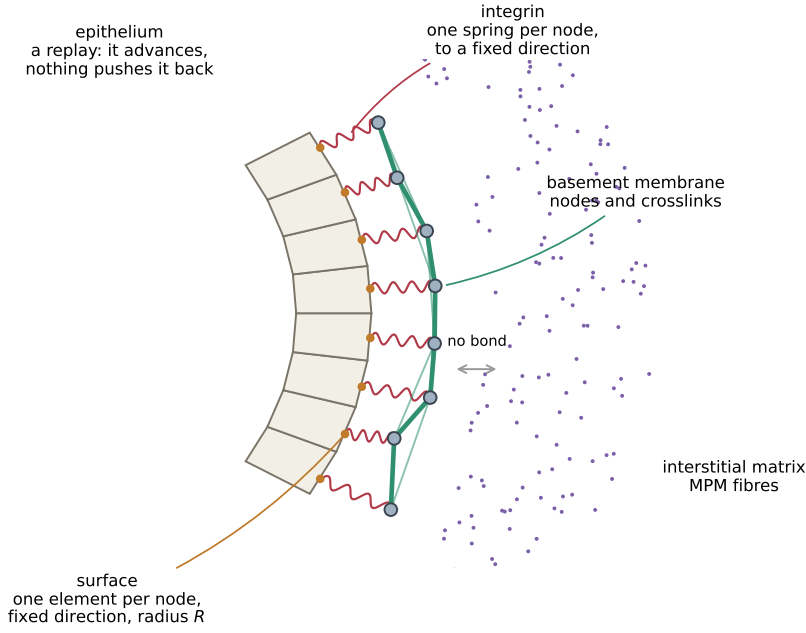


Figure 1: Four objects, and the arrows between them are the model. The epithelium advances on a recorded schedule and nothing pushes it back. Its outer boundary is the **surface** Level: one node per membrane node, each holding a fixed direction and the radius the tissue reached in it. The membrane is grey *nodes* joined by green *crosslinks*, sitting a distance δ outside, each node tied to its own surface node by one red *tether*. The matrix beyond is never bonded to.

6. Junction myosin

In the vertex model every cell–cell contact pulls with the same strength. Real contractility is not uniform: myosin accumulates on some junctions and not others, and where it accumulates the junction pulls harder and shortens, which is how cells exchange neighbours. Giving each junction its own myosin is what lets that happen.

Each junction carries an amount of myosin m_e multiplying the line tension, chasing a target on its own timescale:

$$\Lambda \sum_e \ell_e \longrightarrow \Lambda \sum_e m_e \ell_e, \quad m_e^{ss} = a \left(1 + \beta (x_e / \langle x \rangle - 1) \right), \quad \dot{m}_e = (m_e^{ss} - m_e) / \tau_m. \quad (7)$$

What matters is what x_e is. If it is the junction’s *length*, the loop reads *longer* $\rightarrow$ *more myosin* $\rightarrow$ *shorter*, which evens the cells out and suppresses neighbour exchange. Myosin is in fact recruited by *tension*, and accumulates on junctions in the act of disappearing — the opposite: a junction that pulls harder shortens, which makes it pull harder still. So x_e is the edge tension, $\Lambda m_e + 2K_P(P_f - P_f^0) + \Gamma P_f$, taken from the same energy the mechanics minimises.

The two versions cannot be told apart by the spread of junction lengths, because in the fast-myosin limit — the correct one, since myosin turns over in seconds — substituting the target into Eq. (7) turns the length version into a line tension plus a harmonic spring on each edge, and narrowing that spread is what such a spring does by definition. They differ in how often cells swap neighbours, so that is the observable we report: keyed to tension, the rate of neighbour exchange is roughly double.

Two things the model does not have. The myosin responds to the *magnitude* of tension and has no preferred direction in the plane of the sheet, so it cannot express the planar polarity that drives directional tissue elongation. And τ_m is a chosen number, not a measured one.

7. Where this stops, and what it measured

Sixty-six runs (91–164, archived) were spent asking where the sheet sits and what holds it there, and they answer that question and no other. Four results are worth carrying: the standoff is a *tracking lag*, $\delta - \dot{R} \Delta t \gamma / \kappa$, not a force balance, which two fibre lengths differing fourfold confirmed to 3×10^{-5} box units; a force routed through the grid leaves the sheet’s deformation gradient carrying 100% of the true stretch while the same force integrated by the engine leaves it carrying 13%; an MPM fibre cannot act as a rope, because $\mathbf{F}$ is advected by the grid’s velocity gradient and never measures the distance between two particles, so two ends more than a cell apart are two bodies; and the best configuration reached (153) put the membrane at +0.0011 box units with 2.41 of the 2.43 stretch the geometry implies. What none of them is, is a model derived from what a basement membrane is. The rest of this note starts from that instead.



Figure 2: Two cells share an edge, and that edge *is* the junction — it belongs to neither, which is why its state is keyed by the pair of vertices rather than stored on a cell. Myosin sits on the shared edge and multiplies its tension. Right: the feedback, which is the only thing here the original model cannot already do.

8. Spheroid $\leftrightarrow$ BM $\leftrightarrow$ ECM

Three objects, and every arrow between them is a measured mechanism rather than a modelling choice. The first thing the measurements say is that the middle object is not a passive spacer. A basement membrane is ~ 100 nm thick, its adhesions are *squat* — plaques ~ 350 nm across on ~ 700 nm centres, joined to the cell across a ~ 30 nm gap, so an adhesion is three times wider than the sheet is thick and a third as long — and the sheet is 10 to 100 times stiffer than the stroma outside it, not equal to it. The second thing they say is that it is not a permanent solid: collagen IV in the fly embryo has a half-life of 7–10 hours and in mouse hair follicle about 3, and slowing that turnover disrupts morphogenesis. A sheet enclosing a tissue that triples its radius does not stretch to 2.4; it is *replaced* at the new size while it is being stretched.

Mass, not just mechanics. The sheet carries an areal density ρ of collagen IV that is produced by the epithelium and removed by proteolysis, on a surface whose area $A(t)$ is growing:

$$\frac{D\rho}{Dt} = \underbrace{s(\mathbf{x}, t)}_{\text{secretion}} - \underbrace{\frac{\rho}{\tau_{\text{bm}}}}_{\text{turnover}} - \underbrace{\rho \frac{\dot{A}}{A}}_{\text{dilution by growth}}, \quad \tau_{\text{bm}} = \frac{t_{1/2}}{\ln 2} \approx 4\text{--}14 \text{ h}. \quad (8)$$

Homeostasis is $s^* = \rho(1/\tau_{\text{bm}} + \dot{A}/A)$, and the two terms are comparable here: a spheroid tripling its radius over a day has $\dot{A}/A \sim 10^{-1} \text{ h}^{-1}$ against $1/\tau_{\text{bm}} \sim 10^{-1} \text{ h}^{-1}$. Turnover is not a correction to growth; it is the same size as growth, which is why the sheet can accommodate it at all.

A nonlinear sheet. A single Young's modulus is wrong by construction: the same basement membrane reads ~ 0.4 MPa at low strain and ~ 3 MPa at high strain, and a growing spheroid takes it across that whole range. The cheapest law that respects it makes the tangent modulus rise with the principal stretch $\lambda = \sigma_{\text{max}}(\mathbf{F})$,

$$E(\lambda) = E_0 [1 + \beta (\lambda - 1)], \quad E_0 \approx 0.4 \text{ MPa}, \quad \beta \approx 5, \quad \frac{E_{\text{bm}}}{E_{\text{ecm}}} \in [10, 100]. \quad (9)$$

Adhesion with its reaction, and with slip. An integrin plaque is a two-sided object: whatever force it applies to the sheet it applies equally and oppositely to the cell, which a replayed epithelium currently discards. And the sheet is not pinned — it *slides* relative to the cells — so the tangential law is friction against relative velocity, not a spring to a frozen direction:

$$\mathbf{f}_p^{\text{adh}} = \kappa_n (\ell_p - \ell_0) \hat{\mathbf{n}}_p - \xi (\mathbf{v}_{\text{bm}} - \mathbf{v}_{\text{epi}})_{\parallel}, \quad \mathbf{f}_p^{\text{epi}} = -\mathbf{f}_p^{\text{adh}}, \quad (10)$$

$$n_p = \Sigma^{-2} \text{ plaques per unit area}, \quad \Sigma \approx 7T, \quad \ell_0 \approx 0.3T. \quad (11)$$

ℓ_0 , the linkage's rest length, is what sets the standoff — a material property, not a tuning constant — and ξ is what lets a sheet stay attached while the tissue slides underneath it.

Sheet to stroma. The two are joined by anchoring fibrils, so the stroma is loaded through the sheet and never directly by the cells; the same fibrils are what let a stiff sheet spread a local epithelial push over a soft matrix. With

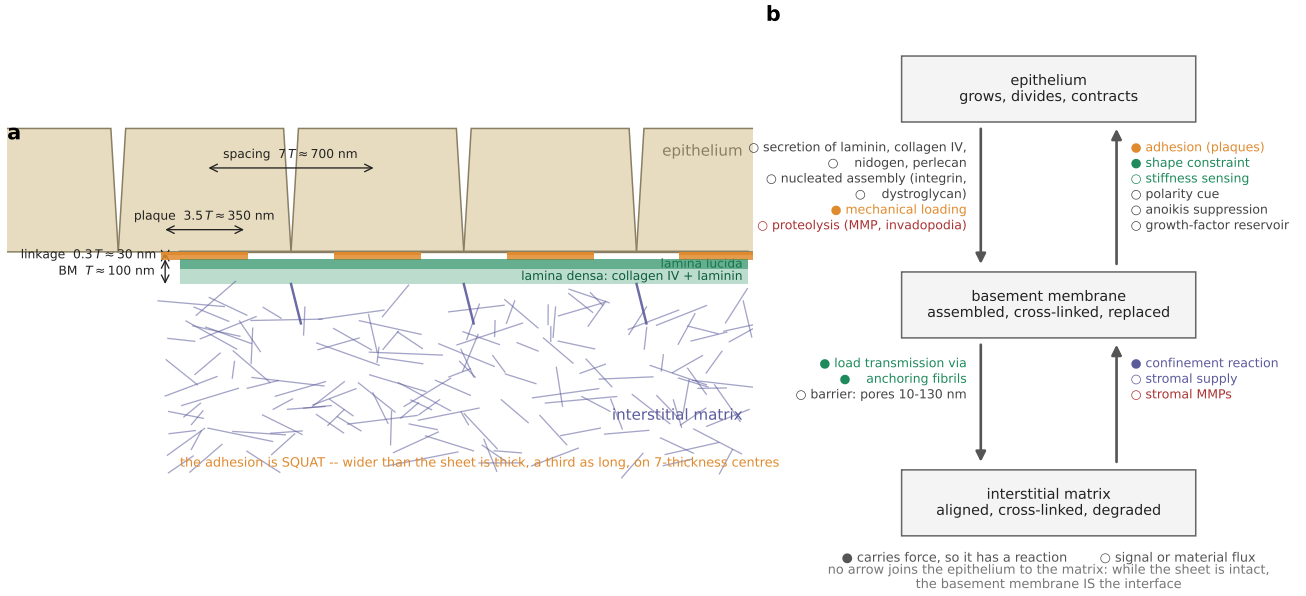


Figure 3: (a) The three entities in section, at the proportions the literature measures rather than the ones that are convenient: the adhesion is a squat plaque, not a thread, and the sheet is thin against its own spacing. (b) Every operator, in the direction it acts. Filled bullets carry a force and must therefore have a reaction; open bullets carry a signal or a material flux. There is no arrow between the epithelium and the matrix, because while the sheet is intact the basement membrane *is* the interface — an epithelium that also pushes the stroma directly is counting the same push twice.

$\mathbf{t}$ the traction the sheet exerts on the stroma,

$$\mathbf{t} = k_a (\mathbf{x}_{\text{ecm}} - \mathbf{x}_{\text{bm}}) \quad \text{on anchored points,} \quad \text{and} \quad \mathbf{f}^{\text{bm}} = - \sum_{\text{anchors}} \mathbf{t}, \quad (12)$$

with the barrier being geometric rather than mechanical: pores of 10–130 nm pass solutes, stop cells, and are what a proteolytic operator has to open before anything crosses.

What this asks of the implementation. Four things, and each is a separate operator with its own test: a mass balance on the sheet (Eq. 8) rather than a fixed particle budget; a strain-dependent modulus (Eq. 9) rather than one number; an adhesion that returns its reaction and permits slip (Eq. 10) rather than a tether to a frozen direction; and an anchorage to the stroma (Eq. 12) that removes the epithelium’s direct push on it.

9. The same thing as a Plexus2 container

Two questions decide whether the biology above can be written down: what the *sets* are, and what kind of operator each mechanism is. Both have answers the literature forces, and both differ from what the archived line built.

Sets. An adhesion is not a body and a fibril is not a particle — each is a *relation between two entities*, which in Plexus2 is an edge-set carrying `.pre` and `.post` index buffers rather than a set of nodes with positions. Writing the hemidesmosome as an edge-set is what makes its reaction automatic: one operator on `plaque` returns a delta to both endpoints, so the force the sheet feels and the force the cell feels cannot get out of step. The archived line made the adhesion first a field on the membrane and then a column of MPM particles, and both lost the reaction.

set	kind	one element is	state it carries
spheroid	node, depth 0	the organ	centroid, cell count
cell	node, depth 1	one epithelial cell	A, P, V and their targets, age
vertex	node, depth 1	a corner of the cell mesh	pos, vel
junction	$\text{edge} \subseteq \text{vertex}^2$	a cell-cell contact	length, myosin
bm	node, depth 1	a patch of basement membrane	$\text{pos}, \mathbf{F}, \rho$ (areal density), age since deposition
plaque	$\text{edge cell} \rightarrow \text{bm}$	one hemidesmosome	rest length ℓ_0 , load, bound flag
ecm	node, depth 1	a patch of interstitial matrix	$\text{pos}, \mathbf{F}$, fibre direction
fibril	$\text{edge bm} \rightarrow \text{ecm}$	one anchoring fibril	rest length, load
mpm_grid	field	momentum on a background lattice	mass, momentum
protease	field	MMP concentration	scalar, diffusing
growth_factor	field	FGF/VEGF bound by perlecan	scalar, bound + free

Operators. One row per mechanism in Fig. 3b, in Plexus2’s own taxonomy: *Lateral* moves state along an edge set, *Exchange* scatters to or gathers from a field, *Rewire* changes which elements are connected, and *Divide/Die* change how many elements exist. The last column is what would falsify the operator — the measurement its test folder has to make.

operator	kind	acts on	mechanism, from the literature	what its test measures
<i>epithelium</i>				
grow_3d, divide_3d	Divide	cell	volume growth and cytokinesis	cells $200 \rightarrow 6076$, radius $\times 3.56$
shape_energy_3d	Lateral	junction	vertex-model force balance	aspect 1.021, stays round
reconnect_t1_3d	Rewire	junction	neighbour exchange	0.0089 T1 per cell per frame
<i>epithelium $\rightarrow$ basement membrane</i>				
bm_secrete	Divide	bm	polarised deposition of laminin, collagen IV, nidogen, perlecan	areal density ρ held constant as $A(t)$ grows
bm_assemble	Rewire	bm	laminin polymerises where integrin and dystroglycan nucleate it	no sheet forms where the receptor is absent
bm_degrade	Die	bm	MMP proteolysis, local and MMP-independent breach	a hole opens only where protease is present
<i>basement membrane $\leftrightarrow$ epithelium</i>				
plaque_seed	Rewire	plaque	hemidesmosomes form at density Σ^{-2} , $\Sigma \approx 7T$	plaque count and spacing
plaque_pull	Lateral	plaque	Eq. (10): normal spring at rest length ℓ_0 , tangential friction ξ , and its reaction	standoff = ℓ_0 ; momentum conserved to machine precision
plaque_rupture	Rewire	plaque	the bond fails past a load	fraction bound vs. applied stress
bm_stiffness	Lateral	bm	Eq. (9): tangent modulus rising with stretch	E $0.4 \rightarrow 3$ MPa across the run
<i>basement membrane $\leftrightarrow$ matrix</i>				
fibril_seed, fibril_pull	Rewire, Lateral	fibril	collagen VII anchoring; the stroma is loaded <i>through</i> the sheet	stroma displacement with the sheet present vs. absent
mpm_scatter/gather	Exchange	bm, ecm	both bodies share one momentum field	contact without an explicit contact law
bm_barrier	Exchange	bm, fields	pores 10–130 nm pass solutes, stop cells	solute flux across an intact vs. breached sheet
<i>matrix</i>				
ecm_align, ecm_crosslink	Lateral	ecm	fibre reorientation under load; LOX stiffening	alignment angle vs. radial direction

Three of these have no counterpart in the archived implementation and are the reason it stalled: **bm_secrete** as a mass balance rather than a fixed particle budget, **plaque_pull** returning its reaction, and **fibril_pull** replacing the epithelium’s direct push on the stroma. Each gets its own folder, spec, movie and measurement plot, in the order of the table.

10. Symbols

Grouped by what they belong to. Where a symbol is a stiffness, the second column says what it is a stiffness of, since several are easy to confuse.

Epithelium — Eqs. (1), (2)		
A_f, P_f, V_f	area, perimeter, volume	of cell (face) f ; A_f^0, P_f^0, V_f^0 are its targets
K_A, K_P, K_V	stiffnesses	how hard a cell resists changing its area / perimeter / volume
Λ	line tension	pull along every cell–cell contact, one value for the whole tissue
Γ	perimeter contractility	a second, quadratic pull on each cell’s perimeter
ℓ_e	length of junction e	
K_R, R_0	radial restoring stiffness, radius	holds the shell against collapse
f_{grow}	per-frame growth factor	applied to a cell’s target volume
Junction myosin — Eq. (7)		
m_e	myosin on junction e	dimensionless multiplier on Λ ; mean 1
x_e	what the feedback reads	the junction’s tension
a	global myosin level	multiplies every junction equally
β	feedback strength	how strongly m_e responds to x_e above average
τ_m	myosin timescale	how fast m_e chases its target
Surface and matrix — Eqs. (3), (4)		
$\hat{\mathbf{u}}_i$	direction	fixed unit vector from the tissue centre $\mathbf{c}$
$R(\hat{\mathbf{u}}, t)$	surface radius	how far the epithelium reached in that direction
r_i	radius of particle i	
k_c	contact stiffness	how hard the surface pushes a fibre back out
P	matrix pressure	contact force per unit surface area, by direction
$p_{1/2}, n, f$	gate half-point, sharpness, floor	g is the resulting growth multiplier, $f \leq g \leq 1$
Basement membrane — Eqs. (5), (6)		
$\mathbf{x}_i$	position of node i	
γ	drag	converts force to speed: $\dot{\mathbf{x}} = \mathbf{F}/\gamma$
k_b	crosslink stiffness	how hard a stretched crosslink pulls its two nodes together
L_{ij}, ℓ_{ij}^0	current, rest length	of the crosslink between nodes i and j
$\hat{\mathbf{d}}_{ij}$	unit vector	from node i toward node j
τ_r	remodelling timescale	how fast ℓ^0 creeps toward L , i.e. how fast it forgets
ε_b	break strain	the stretch at which a crosslink snaps
κ	tether stiffness	how hard a node is pulled back to its anchor (not k_b)
$\mathbf{a}_i$	anchor	the point on the surface node i is tied to
δ	offset	how far outside the surface the sheet sits